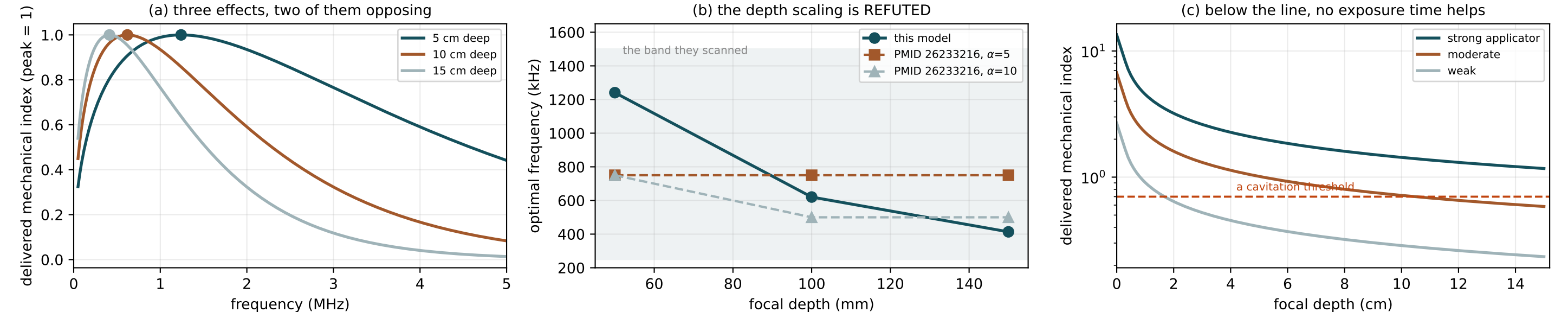


Sonodynamic therapy: an optimum the model finds, and a scaling an independent study contradicts



(a) Focusing favours a high frequency, while attenuation and the mechanical index's own $1/\sqrt{f}$ both favour a low one, so the product peaks in between. (b) Ellens & Hynynen 2015 (PMID 26233216) simulated the same three mechanisms by an independent full-wave route and report a nearflat optimum where this model predicts a 3x fall across 50-150 mm. They name the missing term themselves - near-field heating, which this model has no representation of at all. (c) is why SDT is structurally unlike the dose-response arms: inertial cavitation is a THRESHOLD, so below the line no insonation time produces sonochemical ROS and the depth limit is a hard edge rather than a fade - where each curve crosses the line IS the depth limit tabulated in the validation page. The threshold height is a parameter, not a measurement: published in-vivo values move by more than an order of magnitude with nucleation, and the curves are drawn at a 20 MHz device cap, which is what keeps them finite at the surface: uncapped, this model diverges where nothing has yet attenuated.